Proof that Enhanced Pyramid Networks are 2-Edge-Hamiltonicity

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Abstract

The pyramid network is a hierarchy structure based on meshes. Chen et al. in 2004 proposed a new hierarchy structure, called the enhanced pyramid network, by replacing each mesh at layer greater than one of the pyramid network with a torus. This investigation demonstrates that the enhanced pyramid network with two faulty edges is Hamiltonian. The result is optimal because both edge connectivity and node connectivity of the enhanced pyramid network are four.

1 Introduction

The *pyramid network* (*PM*, for short), is a well-known interconnection network (network, for short) in image processing, computer vision, parallel computing, and network computing [7], [8], [11]. The *PM* is a hierarchy structure based on meshes. Recently, there are many researches on the *PM*, such as Hamiltonicity [4], [10], [18], pancyclicity [4], fault tolerance [2], routing [10], and broadcasting [9]. Note that the node degree of the *PM* is from 3 to 9, and its node connectivity and edge connectivity are both 3 [2], [18]. For establishing the *PM* in expandable VLSI chips, each of its nodes should be configured as a 9-port component or too many different components should be designed and fabricated. In other words, those nodes of degree less than 9 have unused ports. These ports can be used for further expansion or I/O communication. For modifying a well-known network a little bit such that the resulting network has better topological properties, Chen et al. proposed a variant network of the *PM*, the enhanced pyramid network, by connecting some of the unused ports [6].

The *enhanced pyramid network* (*EPM*, for short) is a supergraph of the *PM* with the same node set. In other words, the *PM* is a spanning subgraph of the *EPM*. The *EPM* can be constructed by replacing each mesh at layer greater than one of the *PM* with a torus. Therefore, the hardware cost of the *EPM* would be slightly expensive than the *PM* because some extra edges have to be placed in the VLSI chip. Some topological properties of the *EPM*, including the number of nodes (edges), node connectivity, edge connectivity, and diameter, have been determined [6]. Also, a shortest-path routing algorithm [6] and an optimal broadcasting algorithm [5] have already been derived for the *EPM*. These results show that the *EPM* has better topological properties than the *PM* such as larger node (edge) connectivity and better fault-tolerance ability.

An embedding of one *guest network* G into another *host network* H is a one-to-one mapping m from the node set of G to the node set of H such that an edge of G corresponds to a path of H under m [12]. Clearly, the *PM* can be embedded into the *EPM* and each edge of the *PM* corresponds to an edge of the *EPM*.

A cycle (path) that contains every node of a network is defined to be a *Hamiltonian cycle* (*path*). A connected network having a Hamiltonian cycle is called *Hamiltonian*. The *Hamiltonicity* of a network indicates its capability of embedding cycles (paths). Paths and cycles are two of the most fundamental networks for parallel and distributed computation, and suitable for designing simple algorithms with low communication costs. Various algebraic problems and graph problems can be efficient solved on paths and cycles [1], [12].

Since node or edge faults may happen when a network is put in use, it is very meaningfully to consider the faulty network. Topological properties of the faulty networks were investigated by literatures [2], [3], [13], [15], and [19]. These researches related to the faulty networks include computing diameter, fault diameter, wide diameter, routing, multicasting, broadcasting, embedding rules, and so on. We concentrate our attention on the fault-tolerant embedding cycles in the *EPM*.

The rest of this work is structured as follows. Section 2 describes some notations in graphs and definitions of three operations for the *EPM*. Section 3 first shows that the *EPM* with one faulty node or one faulty edge is Hamiltonian. Then, Section 4 reveals that the 2-layer *EPM* with two faulty edges is Hamiltonian. Section 5 proves that the *EPM* with two faulty edges is Hamiltonian using mathematic induction. Conclusions are finally drawn in Section 6.

2 Preliminaries

The *PM* and *EPM* are now defined. The three operations that are used in the remainder of this paper are also defined.

The node set of a mesh $M(m, n)$ is $V(M(m, n)) = \{(x, y) \mid 0 \le x < m, 0 \le y < n\}$. Two nodes (x_1, y_1) and (x_2, y_2) are joined by an edge iff $|x_1 - x_2| + |y_1 - y_2| = 1$, where (x_1, y_1) and (x_2, y_2) belong to $V(M(m, n))$. The n-layer pyramid network, denoted by $PM[n]$, is a hierarchy structure based on meshes. That is, the $PM[n]$ is a combination of trees and meshes. The node set of $PM[n]$ is $V(PM[n]) = \{(k; x, y) \mid 0 \le k \le n, 0 \le x, y < 2^k\}$, where $n \ge 0$.

The *EPM* is the conjunction of trees and tori. A torus is also called a toroidal mesh [16]. The node set of a torus $T(m, n)$ is $V(T(m, n)) = \{(x, y) \mid 0 \le x < m, 0 \le y < n\}$. Let uv denotes an edge connecting nodes u and v of a network. The edge set of a torus $T(m, n)$ is $E(T(m, n)) = E(M(m, n)) \cup \{(x, 0)(x, n-1), (0, y)(m-1, y) \mid 0 \le x < m, 0 \le y < n\}$. The *EPM* can be constructed by replacing each mesh at layer greater than one of the *PM* with a torus.

For simplicity, let $(a)_b$ denote a mod b. An enhanced pyramid network of n layers is denoted by $EPM[n]$. The node set of $EPM[n]$ is $V(EPM[n]) = \{(k; x, y) \mid 0 \le k \le n, 0 \le x, y < 2^k\}$, where $n \ge 2$. A node $(k; x, y)$ of the *PM* (*EPM*) is said to be a node at layer k, *row* y and *column* x of $M(2^k, 2^k)$ ($T(2^k, 2^k)$) and the nodes at layer k greater than one of the *PM* (*EPM*) are connected as a $M(2^k, 2^k)$ ($T(2^k, 2^k)$). For an edge of the *EPM*, we name it as a *row-edge* (*column-edge*) if both of its endnodes are in row y (column x) for some $0 \le x, y < 2^k$. A node $(k; x, y) \in V(EPM[n])$, $0 < k < n$, is adjacent to $(k-1; \lfloor x/2 \rfloor, \lfloor y/2 \rfloor)$, $(k; (x-1)_{2^k}, y)$, $(k; (x+1)_{2^k}, y)$, $(k; x, (y-1)_{2^k})$, $(k; x, (y+1)_{2^k})$, $(k+1; 2x, 2y)$, $(k+1; 2x+1, 2y)$, $(k+1; 2x, 2y+1)$, and $(k+1; 2x+1, 2y+1)$. We say $(k-1; \lfloor x/2 \rfloor, \lfloor y/2 \rfloor)$ is the *parent* of $(k; x, y)$ and each of $(k+1; 2x, 2y)$, $(k+1; 2x+1, 2y)$, $(k+1; 2x, 2y+1)$, and $(k+1; 2x+1, 2y+1)$ is a *child* of $(k; x, y)$. In general, the node $(0; 0, 0)$ is called the *apex* of $EPM[n]$ (apex, for short). Figure 1 depicts the structure of $EPM[2]$. Node u is a *neighbor* of node v if an edge joins them. All the neighbors of the node $(k; x, y)$ are shown in Figure 2, where $0 < k < n$ and $0 < x, y < 2^k$. Note that the apex has four neighbors only. In other words, the degree of the apex is four. Obviously, the node degree of $EPM[n]$ is from four to nine. The node connectivity and edge connectivity of $EPM[n]$ are both four [6].

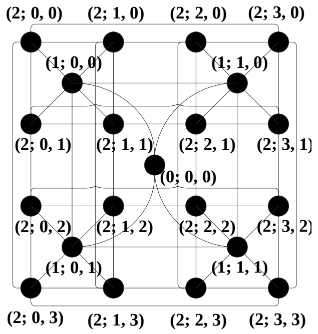


Figure 1. The structure of $EPM[2]$.

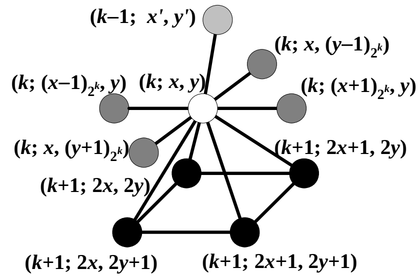


Figure 2. Neighbors of $(k; x, y)$ in $EPM[n]$, where $0 < k < n$, $x' = \lfloor x/2 \rfloor$, and $y' = \lfloor y/2 \rfloor$.

For easy to discuss, we classify the edges of $EPM[n]$ according to the nodes connecting. Let $(k_1; x_1, y_1)(k_2; x_2, y_2)$ be an edge of $EPM[n]$, where $(k_1; x_1, y_1)$ and $(k_2; x_2, y_2)$ belong to $V(EPM[n])$. Without lose of generality, we assume that $k_1 \le k_2$. The edge satisfies one of the following statements:

(1) $k_2 = k_1 + 1$, $x_1 = \lfloor x_2/2 \rfloor$, $y_1 = \lfloor y_2/2 \rfloor$.

(2) $k_2 = k_1$, $|x_1 - x_2| + |y_1 - y_2| = 1$.

(3) $k_2 = k_1$, $|x_1 - x_2| + |y_1 - y_2| = 2^{k_1} - 1$, where $x_1 = x_2$ or $y_1 = y_2$.

We call the edge satisfies (1) a *L-edge* (or *layer-edge*), (2) a *M-edge* (or *mesh-edge*), and (3) a *T-edge* (or *torus-edge*). Moreover, we especially name them L_i*-edge*, M_i*-edge*, and T_i*-edge*, respectively, for $i = k_2$. An *IM-edge* (or *internal mesh-edge*) uv represents that nodes u and v have a common parent. Otherwise, a mesh-edge is an *EM-edge* (or *external mesh-edge*) if its two endnodes have different parents. Similarly, we name them IM_i-edge and EM_i-edge if they are also M_i-edges.

In order to illustrate easily, we will use three operations for $EPM[n]$ in the rest of this paper. We state them as follows:

(1) *Rotation*, denoted by $R_\theta(G)$: Clockwise rotate $EPM[n]$ with degree θ, where $\theta = 90°$, 180°, or 270°. If we clockwise rotate $EPM[n]$ with 90°, 180°, or 270°, then the node $(k; x, y)$ in $EPM[n]$ becomes $(k; 2^k–y–1, x)$, $(k; 2^k–x–1, 2^k–y–1)$, or $(k; y, 2^k–x–1)$, respectively.

(2) *Flipping*, denoted by $F_L(G)$: Flip the *EPM* according to *L*-line through the apex, where *L* represent / (anti-diagonal), \ (diagonal), | (vertical), or — (horizontal). If we flip $EPM[n]$ according to /, \, |, or —, then the node $(k; x, y)$ in $EPM[n]$ becomes $(k; 2^k–y–1, 2^k–x–1)$, $(k; y, x)$, $(k; 2^k–x–1, y)$, or $(k; x, 2^k–y–1)$, respectively.

(3) *Exchange*, denoted by $X_L(G)$: Exchange $EPM[n]$ according to *L*-line through the apex, where *L* represent | (vertical), or — (horizontal). If we exchange $EPM[n]$ according to | or —, then the node $(k; x, y)$ in $EPM[n]$ becomes $(k; (2^{k–1}+x)_{2^k}, y)$, or $(k; x, (2^{k–1}+y)_{2^k})$, respectively.

Note that after any operation describing above the resulting network is still an $EPM[n]$. Any M_1-edge of $EPM[n]$ would become (1; 0, 0)(1; 1, 0) after applying $R_\theta(EPM[n])$. Any IM_2-edge of the *EPM* would become (2; 0, 0)(2; 1, 0) after applying a composition of $X_|(EPM[n])$, $X_—(EPM[n])$, $F_\backslash(EPM[n])$, $F_/(EPM[n])$, and/or $R_\theta(EPM[n])$. Any EM_2-edge of $EPM[n]$ would become (2; 1, 1)(2; 2, 1) after applying a composition of $X_—(EPM[n])$ and/or $R_\theta(EPM[n])$. Any L_1-edge of $EPM[n]$ would become (0; 0, 0)(1; 1, 0) after applying $R_\theta(EPM[n])$. Any L_2-edge of $EPM[n]$ would become (1; 0, 0)(2; 0, 0) after applying a composition of $X_|(EPM[n])$, $X_—(EPM[n])$, and/or $R_\theta(EPM[n])$. This is a very interesting property such that we can only focus on some edges for edge-fault computation and communication on $EPM[2]$ even on $EPM[n]$.

3 1-Hamitonicity of *EPM*[*n*]

In [17], the performance of the Hamiltonian property in the faulty networks is discussed. When faults occur in a network, edges and/or nodes would be removed immediately. Let $G = (V, E)$ be a loopless undirected graph, and $V' \subseteq V$, $E' \subseteq E$. We use $G – V'$ to denote the subgraph of G induced by $V – V'$, and $G – E'$ to denote the subgraph obtained by removing E' from G. A graph G is *f-Hamiltonicity* if $G – F$ is Hamiltonian for every faulty set $F \subset V \cup E$ with $|F| = f \le \delta(G) – 2$ and an *f-Hamiltonian cycle* of G is a Hamiltonian cycle in $G – F$, where $\delta(G)$ denotes the minimum degree of G. If the faulty set F is a subset of the node set of G, G is *f-node-Hamiltonicity*. Similarly, G is *f-edge-Hamiltonicity* if the faulty set F is a subset of the edge set of G. Notice that in this paper a network and a graph are interchangeable.

We have already known that $PM[n]$ with one fault is still Hamiltonian [18]. Since the apex of $EPM[n]$ only has four neighbors, the minimum degree of $EPM[n]$ is four. Therefore, we concentrate on studying whether $EPM[n]$ is also Hamiltonian with two faults. Firstly, we consider 1-Hamiltonicity in $EPM[n]$.

Lemma 1 [18]. $PM[n]$ is 1-Hamiltonicity.

Lemma 2. $EPM[n]$ is 1-node-Hamiltonicity.

Proof. Because $PM[n]$ is a spanning subgraph of $EPM[n]$, the Hamiltonian cycle in $PM[n]$ having a faulty node is also a Hamiltonian cycle of $EPM[n]$ with the same faulty node. Therefore, according to Lemma 1, $EPM[n]$ is 1-node-Hamiltonicity. ■

Lemma 3. $EPM[n]$ is 1-edge-Hamiltonicity.

Proof. Due to $PM[n]$ is a spanning subgraph of $EPM[n]$, $PM[n]$ is faultless when the faulty edge is a *T*-edge. If the faulty edge is not a *T*-edge, $PM[n]$ has one faulty edge. Therefore, according to Lemma 1, $EPM[n]$ is 1-edge-Hamiltonicity. ■

By Lemmas 2 and 3, we have the following theorem.

Theorem 1. $EPM[n]$ is 1-Hamiltonicity.

The 2-Hamiltonicity problem on $EPM[n]$ can be divided into three subproblems:

(1) 2-edge- Hamiltonicity,
(2) 1-edge-and-1-node- Hamiltonicity, and
(3) 2-node-Hamiltonicity.

In this paper, we focus on investigating 2-edge-Hamltonicity on $EPM[n]$.

We derive $EPM[n]$ is 2-edge-Hamiltonicity by mathematic induction on layer n. Firstly, we show that $EPM[2]$ is 2-edge-Hamiltonicity as an induction base in the following section.

4 2-Edge-Hamitonicity of *EPM*[2]

This section demonstrates that $EPM[2]$ is 2-edge-Hamiltonicity. For simplicity, let $HC(G)$ ($HP(G)$) denote a Hamiltonian cycle (Hamiltonian path) in a network G, and $HC(G – F)$ denote a Hamiltonian cycle in a network G with faulty set F. Let e_1 and e_2 be two faulty edges in $EPM[2]$. For constructing an $HC(EPM[2] – \{e_1, e_2\})$, we first construct an $HC(T(2^2, 2^2) – F_2)$ and an

$HC(PM[1] - F_1)$, where F_1 and F_2 are two faulty set of edges such that $F_1 \cup F_2 = \{e_1, e_2\}$ and $F_1 \cap F_2 = \emptyset$. Notice that F_1 (F_2) is a subset of the edge set of $PM[1]$ ($T(2^2, 2^2)$). Then we combine them to build a $HC(EPM[2] - \{e_1, e_2\})$ by adding two layer-edges and removing two mesh-edges. For easy to understand the connection of the $HC(T(2^2, 2^2) - F_2)$ and $HC(PM[1] - F_1)$, the coordinate of each node in $HC(T(2^2, 2^2))$ and $PM[1]$ is assigned the same as the coordinate of each node in $EPM[2]$. Since constructing an $HC(PM[1] - F_1)$ is very simple, we only describe how to construct an $HC(T(2^2, 2^2) - F_2)$. We now aim to establish an $HC(T(2^k, 2^k))$. To modify the $HC(T(2^2, 2^2))$ to an $HC(T(2^2, 2^2) - F_2)$ will be described in the proof of Lemma 4. In the following, we derive two algorithms to establish an $HC(T(2^k, 2^k))$, where $k \geq 2$.

Algorithm 1. {//Constructing an $HC(T(2^k, 2^k))$
/* Input: the layer $k \geq 2$
Output: an $HC(T(2^k, 2^k))$ */

Step 1. For each row in $T(2^k, 2^k)$, construct a cycle containing all nodes of it.

Step 2. For $0 \leq i < 2^{k-1}$, add the edges $(k; 0, 2i)(k; 0, 2i+1)$ and $(k; 1, 2i)(k; 1, 2i+1)$, and remove the edges $(k; 0, 2i)(k; 1, 2i)$ and $(k; 0, 2i+1)(k; 1, 2i+1)$.

Step 3. For $0 \leq i < 2^{k-1}-1$, add the edges $(k; 2^k-2, 2i+1)(k; 2^k-2, 2i+2)$ and $(k; 2^k-1, 2i+1)(k; 2^k-1, 2i+2)$, and remove the edges $(k; 2^k-2, 2i+1)(k; 2^k-1, 2i+1)$ and $(k; 2^k-2, 2i+2)(k; 2^k-1, 2i+2)$.

} //End of Algorithm 1

Algorithm 2. {//Constructing an $HC(T(2^k, 2^k))$
/* Input: the layer $k \geq 2$
Output: an $HC(T(2^k, 2^k))$ */

Step 1. For each row in $T(2^k, 2^k)$, construct a cycle containing all nodes of it.

Step 2. For $0 \leq i < 2^{k-1}$, add the edges $(k; 2^k-2, 2i)(k; 2^k-2, 2i+1)$ and $(k; 2^k-1, 2i)(k; 2^k-1, 2i+1)$, and remove the edges $(k; 2^k-2, 2i)(k; 2^k-1, 2i)$ and $(k; 2^k-2, 2i+1)(k; 2^k-1, 2i+1)$.

Step 3. For $0 \leq i < 2^{k-1}-1$, add the edges $(k; 0, 2i+1)(k; 0, 2i+2)$ and $(k; 1, 2i+1)(k; 1, 2i+2)$, and remove the edges $(k; 0, 2i+1)(k; 1, 2i+1)$ and $(k; 0, 2i+2)(k; 1, 2i+2)$.

} //End of Algorithm 2

In Step 1 of Algorithms 1 and 2, we first construct 2^k separated cycles of length 2^k in $T(2^k, 2^k)$. Step 2 combines $2i^{\text{th}}$ and $(2i+1)^{\text{th}}$ cycles to from 2^{k-1} separated cycles of length 2^{k+1}. Step 3 merges the 2^{k-1} cycles of length 2^{k+1} to establish an $HC(T(2^2, 2^2))$. Clearly, the constructed $HC(T(2^k, 2^k))$ must contain all T-edges that are also row-edges. Note that the $HC(T(2^k, 2^k))$ constructed by Algorithm 2 is an alternative to the one by Algorithm 1. Figure 3 illustrates the $HC(T(2^2, 2^2))$ constructed by Algorithm 1.

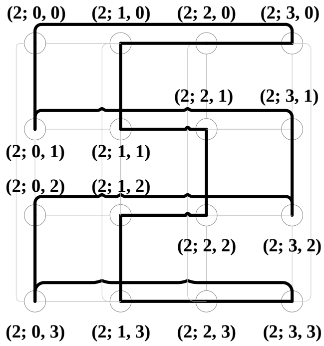


Figure 3. The $HC(T(2^2, 2^2))$ constructed by Algorithm 1.

Lemma 4. *EPM*[2] is 2-edge-Hamiltonicity.

Proof. $PM[n]$ is Hamiltonian [14], [18] and 1-Hamiltonicity [18], and $PM[2]$ is a spanning subgraph of $EPM[2]$. If at most one faulty edges is not a T-edge, there is at most one faulty edge in a spanning subgraph $PM[2]$ of $EPM[2]$. The Hamiltonian cycle of $PM[2]$ is also a Hamiltonian cycle of $EPM[2]$. In the following discussion, we concentrate our attention on two faulty edges which are both not T-edges.

Let e_1 and e_2 be two faulty edges but not T-edges in $EPM[2]$. We first construct an $HC(T(2^2, 2^2) - F_2)$ and an $HC(PM[1] - F_1)$ containing a mesh-edge $(1; 0, 0)(1; 1, 0)$, where F_1 and F_2 are two faulty set of edges such that $F_1 \cup F_2 = \{e_1, e_2\}$ and $F_1 \cap F_2 = \emptyset$. Then we combine them to form a $HC(EPM[2] - \{e_1, e_2\})$ by adding two layer-edges $(1; 0, 0)(2; 1, 1)$ and $(1; 1, 0)(2; 2, 1)$, and removing two mesh-edges $(1; 0, 0)(1; 1, 0)$ and $(2; 1, 1)(2; 2, 1)$. Only the faulty edge(s) containing in $HC(T(2^2, 2^2))$ as shown in Figure 3 have to be concerned for establishing it. And we do not care about the edge $(2; 1, 1)(2; 2, 1)$ because it is not in $HC(T(2^2, 2^2) - F_2)$. Three cases should be considered according to the type of the two faulty edges, described as follows.

Case 1. (Both two faulty edges are M-edges): Since anyone of M_1-edges, IM_2-edges and EM_2-edges of $EPM[2]$ would become $(1; 0, 0)(1; 1, 0)$, $(2; 0, 0)(2; 1, 0)$ or $(2; 1, 1)(2; 2, 1)$ after applying a composition of some proper operation(s), we can only focus on some edges for edge-fault computation. Two subcases should be discussed according to the types of M-edges of $EPM[2]$:

Subcase 1.1. (At most one faulty edge is an M_2-edge): Without lose of generality, we assume that e_2 is (2; 0, 0)(2; 1, 0) or (2; 1, 1)(2; 2, 1) if there exists a faulty M_2-edge, and (1; 0, 0)(1, 1, 0) if there is no faulty M_2-edge. Note that the edge (2; 0, 0)(2; 1, 0) is not contained in the $HC(T(2^2, 2^2))$, and the edges (2; 1, 1)(2; 2, 1) and (1; 0, 0)(1, 1, 0) will be removed later. Now we only have to consider the other faulty edge e_1. If e_1 is (1; 1, 0)(1; 1, 1), we apply $X_|(EPM[2])$ such that the modified e_2 is not in the $HC(T(2^2, 2^2))$ except the edge (2; 1, 1)(2; 2, 1) and e_1 is not (1; 1, 0)(1; 1, 1). Firstly, we establish an $HC(PM[1] - \{(1; 0, 1)\})$ as (1; 0, 0) → (0; 0, 0) → (1; 1, 1) → (1; 1, 0) → (1; 0, 0). Then we connect the $HC(PM[1] - \{(1; 0, 1)\})$ and the $HC(T(2^2, 2^2))$ by adding edges (1; 0, 0)(2; 1, 1) and (1; 1, 0)(2; 2, 1) after removing the edge (2; 1, 1)(2; 2, 1) of $HC(T(2^2, 2^2))$. However, the node (1; 0, 1) is not contained in the resulting cycle. We include it by removing the edge (2; 0, 2)(2; 0, 3) and adding two edges (2; 0, 2)(1; 0, 1) and (1; 0, 1)(2; 0, 3). Consequently, a $HC(EPM[2] - \{e_1, e_2\})$ can be obtained.

Subcase 1.2. (Both two faulty edges are M_2-edges):

Subcase 1.2.1. (Both two faulty edges are IM_2-edges): Without loss of generality, we assume that e_2 is (2; 0, 0)(2; 1, 0). Now we only have to discuss that e_1 is an IM_2-edge of the $HC(T(2^2, 2^2))$. We apply $X\_(EPM[2])$ if e_1 is (2; 2, 0)(2; 3, 0) or (2; 2, 3)(2; 3, 3), and apply $X\_(EPM[2])$ and $X_|(EPM[2])$ if e_1 is (2; 0, 0)(2; 0, 1), (2; 1, 0)(2; 1, 1), (2; 0, 2)(2; 0, 3), or (2; 1, 2)(2; 1, 3). After some proper operation(s), the two modified faulty M_2-edges are not in the $HC(T(2^2, 2^2))$.

Subcase 1.2.2. (At least one faulty edge is an EM_2-edge): Without loss of generality, we assume that e_2 is (2; 1, 1)(2; 2, 1). Now we only need to consider that e_1 is an M_2-edge in the $HC(T(2^2, 2^2))$. We apply $R_{90°}(EPM[2])$ and $X\_(EPM[2])$ if e_1 is (2; 1, 0)(2; 2, 0), (2; 1, 2)(2; 2, 2), or (2; 1, 3)(2; 2, 3), and apply $F_|(EPM[2])$ if e_1 is (2; 2, 0)(2; 3, 0), (2; 2, 3)(2; 3, 3), (2; 0, 0)(2; 0, 1), (2; 1, 0)(2; 1, 1), (2; 2, 1)(2; 2, 2), (2; 3, 1)(2; 3, 2), (2; 0, 2)(2; 0, 3), or (2; 1, 2)(2; 1, 3). The two modified faulty M_2-edges are not in the $HC(T(2^2, 2^2))$ except the edge (2; 1, 1)(2; 2, 1).

An $HC(PM[1])$ is constructed as (1; 0, 0) → (0; 0, 0) → (1; 0, 1) → (1; 1, 1) → (1; 1, 0) → (1; 0, 0). Note that the $HC(PM[1])$ and $HC(T(2^2, 2^2))$ must include the two edges (1; 0, 0)(1; 1, 0) and (2; 1, 1)(2; 2, 1), respectively. Finally, we connect the $HC(PM[1])$ and $HC(T(2^2, 2^2))$ by adding two L-edges (1; 0, 0)(2; 1, 1) and (1; 1, 0)(2; 2, 1), and removing two M-edges (1; 0, 0)(1; 1, 0) and (2; 1, 1)(2; 2, 1). Consequently, a $HC(EPM[2] - \{e_1, e_2\})$ can be obtained.

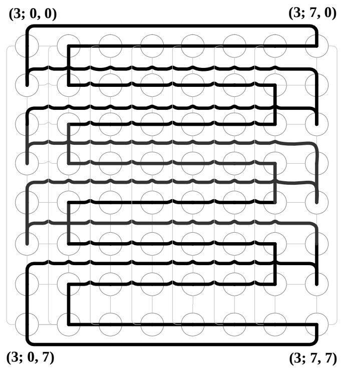
Case 2. (One faulty edge is an M-edge, and the other is an L-edge): Any M_1-edge, IM_2-edges and EM_2-edges of the EPM would become (1; 0, 0)(1; 1, 0), (2; 0, 0)(2; 1, 0) and (2; 1, 1)(2; 2, 1) after applying a composition of some proper operation(s), respectively. Without lose of generality, we assume that e_2 is the faulty M-edge (1; 0, 0)(1; 1, 0) if e_2 is M_1-edge, (2; 0, 0)(2; 1, 0) if e_2 is IM_2-edges, and (2; 1, 1)(2; 2, 1) if e_2 is EM_2-edges. Then e_2 is not in the $HC(T(2^2, 2^2))$ except the edge (2; 1, 1)(2; 2, 1). Now we only have to consider the other faulty L-edge e_1. If e_1 is (1; 0, 0)(2; 1, 1) or (1; 1, 0)(2; 2, 1), we apply a composition of $R_{90°}(EPM[2])$, $X\_(EPM[2])$ and $X_|(EPM[2])$ such that the modified e_2 is not in the $HC(T(2^2, 2^2))$ and he modified e_1 is not (1; 0, 0)(2; 1, 1) and (1; 1, 0)(2; 2, 1). We establish an $HC(PM[1] - F_1)$ as (1; 0, 0) → (0; 0, 0) → (1; 0, 1) → (1; 1, 1) → (1; 1, 0) → (1; 0, 0) if e_1 is (0; 0, 0)(1; 0, 1) or (0; 0, 0)(1; 1, 1), or the modified e_2 is (1; 0, 0)(1; 0, 1). Otherwise, an $HC(PM[1] - F_1)$ is constructed as (1; 0, 0) → (1; 0, 1) → (1; 1, 1) → (0; 0, 0) → (1; 1, 0) → (1; 0, 0). Finally, we establish an $HC(EPM[2] - \{e_1, e_2\})$ by merging the $HC(PM[1] - F_1)$ and the $HC(T(2^2, 2^2))$ as Subcase 1.2.

Case 3. (Both two faulty edges are L-edges): After applying a composition of some proper operation(s), any L_1-edge and L_2-edge of $EPM[n]$ would become (0; 0, 0)(1; 1, 0) and (1; 0, 0)(2; 0, 0), respectively. Without lose of generality, we assume that e_2 is (0; 0, 0)(1; 1, 0) if e_2 is L_1-edge, (1; 0, 0)(2; 0, 0) if e_2 is L_2-edges. Now we only have to consider the other faulty L-edge e_1. If e_2 and e_1 are (0; 0, 0)(1; 1, 0) and (0; 0, 0)(1; 0, 1) respectively, we establish an $HC(EPM[2] - \{e_1, e_2\})$ as Subcase 1.1. Otherwise, let e_1 be the faulty L-edge in Case 2, and we establish an $HC(EPM[2] - \{e_1, e_2\})$ as Case 2.

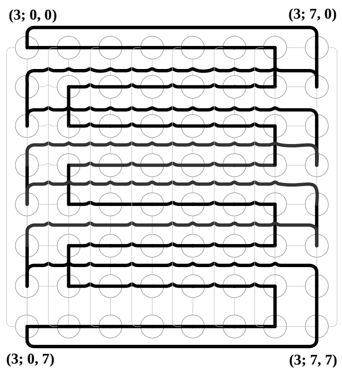
Notice that the resulting $HC(EPM[2] - \{e_1, e_2\})$ contains the two T_2-edges that are 2nd and 3rd rows respectively. ■

5 2-Edge-Hamitonicity of *EPM*[*n*]

Let e_1 and e_2 be two faulty edges in $EPM[n]$. For constructing a $HC(EPM[n] - \{e_1, e_2\})$, we first modify each $HC(T(2^k, 2^k) - F_k)$ established by Algorithm 1 in Section 4, and then combine them with resulting $HC(EPM[2] - F_2)$ in Section 4 to form a $HC(EPM[n] - \{e_1, e_2\})$, where F_k is a faulty set of edges such that $\cup_{2\le k\le n} F_k \subseteq \{e_1, e_2\}$ and $\cap_{2\le k\le n} F_k = \emptyset$. Two $HC(T(2^3, 2^3))$s constructed by Algorithms 1 and 2 are shown in Figure 4(a) and Figure 4(b), respectively. For easy to understand the connection of $HC(T(2^k, 2^k) - F_k)$s and $HC(EPM[2] - F_2)$, the coordinate of each node in $HC(T(2^k, 2^k))$ and $EPM[2]$ is assigned the same as the coordinate of each node in $EPM[n]$.



(a)



(b)

Figure 4. Two $HC(T(2^3, 2^3))$s constructed by (a) Algorithm 1 and (b) Algorithm 2, respectively.

Let the $HC(T(2^k, 2^k))$ constructed by Algorithms 1 and 2 are H_1 and H_2, respectively. We modify H_1 as a $HC(T(2^k, 2^k) - F_k)$ by Algorithms 3 and 4 for $|F_k| = 1$ and 2 respectively, where $3 \le k \le n$. We derive Algorithm 3 to modify the H_1 as a $HC(T(2^k, 2^k) - F_k)$ with $|F_k| = 1$ firstly, where $3 \le k \le n$. The faulty edges contained in the H_1 should be considered.

Algorithm 3. {//Constructing an $HC(T(2^k, 2^k) - \{e_1\})$

/* Input: the layer $k \ge 3$, a faulty edge $e_1 = (k; x_1, y_1)(k; x_2, y_2)$, and an $HC(T(2^k, 2^k))$ H_1
Output: an $HC(T(2^k, 2^k) - \{e_1\})$ */

Step 1. If e_1 is a column-edge in $T(2^k, 2^k)$, H_1 is replaced by H_2, and H_2 is an $HC(T(2^k, 2^k) - \{e_1\})$.

Step 2. If e_1 is a row-edge in $T(2^k, 2^k)$, two cases should be discussed as follows:

Case 1. (e_1 is $(k; 2^k{-}2, 0)(k; 2^k{-}1, 0)$ or $(k; 2^k{-}2, 2^k{-}1)(k; 2^k{-}1, 2^k{-}1)$): H_1 is replaced by H_2, and H_2 is an $HC(T(2^k, 2^k) - \{e_1\})$.

Case 2. (e_1 is $(k; x_1, y)(k; x_2, y)$ but not $(k; 2^k{-}2, 0)(k; 2^k{-}1, 0)$) and $(k; 2^k{-}2, 2^k{-}1)(k; 2^k{-}1, 2^k{-}1)$, where $0 \le y < 2^k$): Let $z = y$ ($z = y-1$) if y is even (odd). An $HC(T(2^k, 2^k) - \{e_1\})$ is established by adding four edges $(k; 0, z)(k; 1, z)$, $(k; 0, z+1)(k; 1, z+1)$, $(k; x_1, z)(k; x_1, z+1)$ and $(k; x_2, z)(k; x_2, z+1)$ after removing four edges $(k; 0, z)(k; 0, z+1)$, $(k; 1, z)(k; 1, z+1)$, $(k; x_1, z)(k; x_2, z)$ and $(k; x_1, z+1)(k; x_2, z+1)$.

} //End of Algorithm 3

In Step 1 of Algorithm 3, since all column-edges of the H_1 are not contained in the H_2, we can establish $HC(T(2^k, 2^k) - \{e_1\})$ as H_2. It is similar in the Case 2 of Step 2. Due to the two row-edges $(k; 2^k{-}2, 0)(k; 2^k{-}1, 0)$ and $(k; 2^k{-}2, 2^k{-}1)(k; 2^k{-}1, 2^k{-}1)$ are not contained in the H_2, we can establish $HC(T(2^k, 2^k) - \{e_1\})$ as H_2. In Case 2 of Step2, we first remove two column-edges and two row-edges (including e_1) in H_1, and then connect these four paths by adding two row-edges and two column-edges which are not in H_1 to build an $HC(T(2^k, 2^k) - \{e_1\})$. Notice that the resulting $HC(T(2^k, 2^k) - \{e_1\})$ by Algorithm 3 still contain all T-edges that are also row-edges.

In the following, we derive Algorithm 4 to modify the H_1 as a $HC(T(2^k, 2^k) - F_k)$ with $|F_k| = 2$, where $3 \le k \le n$. The faulty edges contained in the H_1 should be considered.

Algorithm 4. {//Constructing an $HC(T(2^k, 2^k) - \{e_1, e_2\})$

/* Input: the layer $k \ge 3$, two faulty edges $e_1 = (k; x_1, y_1)(k; x_2, y_2)$ and $e_2 = (k; x_3, y_3)(k; x_4, y_4)$, and an $HC(T(2^k, 2^k))$ H_1
Output: an $HC(T(2^k, 2^k) - \{e_1, e_2\})$ */

Step 1. If both e_1 and e_2 are column-edges in $T(2^k, 2^k)$, two cases should be discussed as follows:

Case 1. (Both e_1 and e_2 are column-edges in H_1): H_1 is replaced by H_2, and H_2 is an $HC(T(2^k, 2^k) - \{e_1, e_2\})$.

Case 2. (Only e_1 is column-edge in H_1): Let e_1 and e_3 be two edges $(k; 1, 2i)(k; 1, 2i+1)$ and $(k; 0, 2i)(k; 0, 2i+1)$ in H_1 respectively, where $0 \le i < 2^{k-1}$. Then choose two edges $e_4 = (k; x, 2i)(k; x, 2i+1)$ and $e_5 = (k; x+1, 2i)(k; x+1, 2i+1)$ which are not in H_1 such that $e_4 \neq e_2$ and $e_5 \neq e_2$, where $2 \le x < 2^k - 2$. An $HC(T(2^k, 2^k) - \{e_1, e_2\})$ is established by adding e_4, e_5, $(k; 0, 2i)(k; 1, 2i)$, and $(k; 0, 2i+1)(k; 1, 2i+1)$ after removing e_1, e_3, $(k; x, 2i)(k; x+1, 2i)$, and $(k; x, 2i+1)(k; x+1, 2i+1)$.

Step 2. If both e_1 and e_2 are row-edges in $T(2^k, 2^k)$, e_1 and e_2 would become two column-edges after applying $R_{90°}(EPM[n])$. If both the two column-edges are not contained in the H_1, the H_1 is also an $HC(T(2^k, 2^k) - \{e_1, e_2\})$. Otherwise, an $HC(T(2^k, 2^k) - \{e_1, e_2\})$ can be constructed as Step 1.

Step 3. If e_1 is a row-edge and e_2 is a column-edge in $T(2^k, 2^k)$, two cases should be discussed as follows:

Case 1. (e_1 is $(k; x_1, y)(k; x_2, y)$ in H_1): If $y \equiv 0 \pmod{2^k}$, apply $X_{-}(EPM[n])$ such that e_1 would become an edge $(k; x_1, y)(k; x_2, y)$, where $y = 2^{k-1}$ or $2^{k-1} - 1$. If both e_1 and e_2 are not in H_1, H_1 is also an $HC(T(2^k, 2^k) - \{e_1, e_2\})$. If e_2 is contained in H_1, apply $X_{|}(EPM[n])$ such that e_2 will not be (e_1 might be) in H_1. If e_1 is still in H_1, an $HC(T(2^k, 2^k) - \{e_1, e_2\})$ is constructed as follows. If e_2 is not $(k; x_1, y)(k; x_1, y+1)$ or $(k; x_2, y)(k; x_2, y+1)$, choose two edges $e_3 = (k; x, y)(k; x, y+1)$ and $e_4 = (k; x+1, y)(k; x+1, y+1)$ which are not in H_1. An $HC(T(2^k, 2^k) - \{e_1, e_2\})$ is constructed by adding $(k; x, y)(k; x+1, y)$, $(k; x, y+1)(k; x+1, y+1)$, $(k; x_1, y)(k; x_1, y+1)$, and $(k; x_2, y)(k; x_2, y+1)$ after removing e_3, e_4, e_1, and $(k; x_1, y+1)(k; x_2, y+1)$. Otherwise, if e_2 is either $(k; x_1, y)(k; x_1, y+1)$ or $(k; x_2, y)(k; x_2, y+1)$, an $HC(T(2^k, 2^k) - \{e_1, e_2\})$ is constructed as above except replacing all $y + 1$ with $y - 1$.

Case 2. (Only e_2 is in H_1): Applying $X_{|}(EPM[n])$, then e_2 will not (but e_1 might be) in H_1. If e_1 is in H_1, an $HC(T(2^k, 2^k) - \{e_1, e_2\})$ is constructed as Case 1.

} //End of Algorithm 4

In Step 1 of Algorithm 4, since all column-edges of the H_1 are not contained in the H_2, we can establish $HC(T(2^k, 2^k) - \{e_1, e_2\})$ as H_2 in Case 1. In Case 2, since both e_1 and e_2 are column-edges in $T(2^k, 2^k)$, e_1 would become $(k; 1, 2i)(k; 1, 2i+1)$ after applying a composition of some proper operation(s), let e_1 be the edge $(k; 1, 2i)(k; 1, 2i+1)$ in H_1. And e_3 is an edge of H_1 decided by the coordinated of e_1. For $k \ge 3$, there are at least $2^k - 4$ choices to choose two edges e_4 and e_5 which are not in H_1 such that $e_4 \neq e_2$ and $e_5 \neq e_2$. We first remove e_1, e_3, and two incident edges of e_4 and e_5, and then connect these four paths by adding two incident edges of e_1 and e_3, e_4, and e_5 to build an $HC(T(2^k, 2^k) - \{e_1, e_2\})$. In Step 2, we apply $R_{90°}(EPM[n])$ such that two row-edges become two row-edges. And $HC(T(2^k, 2^k) - \{e_1, e_2\})$ is established as Step 1. Step 3 discuss that one faulty edge is a row-edge and the other is a column-edge. Firstly, we assume that e_1 is not in row 0 or $2^k - 1$ by applying $X_{-}(EPM[n])$. Then, we determine whether both e_1 and e_2 are in H_1. If not, H_1 is an $HC(T(2^k, 2^k) - \{e_1, e_2\})$. Otherwise, we apply $X_{|}(EPM[n])$ if e_2 is in H_1 such that e_2 will not be (e_1 might be) in H_1. If e_1 is still in H_1, we remove two column-edges and two row-edges (including e_1) in H_1, and then connect these four paths by adding two row-edges and two column-edges which are not in H_1 to build an $HC(T(2^k, 2^k) - \{e_1, e_2\})$.

Notice that the resulting $HC(T(2^k, 2^k) - \{e_1, e_2\})$ still contain all T-edges that are also row-edges. With the aid of Algorithms 1, 2, 3, and 4, we can now show that $EPM[n]$ is 2-edge-Hamiltonicity.

Theorem 2. *EPM*[*n*] is 2-edge-Hamiltonicity.

Proof. We have already known that $PM[n]$ is Hamiltonian [14], [18], 1-Hamiltonicity and $PM[n]$ is a spanning subgraph of $EPM[n]$. If at most one faulty edges is not a T-edge, there is at most one faulty edge in a spanning subgraph $PM[n]$ of $EPM[n]$. The Hamiltonian cycle of $PM[n]$ is also a Hamiltonian cycle of $EPM[n]$. In the following discussion, we concentrate out attention on two faulty edges which are both not T-edges.

Let e_1 and e_2 be two faulty edges but not T-edges in $EPM[n]$. We prove this theorem by mathematic induction on layer n. If $n = 2$, Lemma 4 has shown that $EPM[2]$ is 2-edge-Hamiltonicity and the constructed $HC(EPM[2] - \{e_1, e_2\})$ contains the two T_2-edges that are in 2^{nd} and 3^{rd} rows, respectively. For $n = 2$ to $k - 1$, suppose that $EPM[n]$ is 2-edge-Hamiltonicity and the constructed $HC(EPM[n] - \{e_1, e_2\})$ contains the two T_n-edges that are in the 2^{nd} and $(2^n-1)^{\text{th}}$ rows, respectively. Now, we discuss the condition of $n = k$. Three cases should be considered as follows:

Case 1. (Two faulty edges in $EPM[k-1]$): By induction hypothesis, we have an

$HC(EPM[k–1] – \{e_1, e_2\})$ and an $HC(T(2^k, 2^k))$ constructed by Algorithm 1. We use two L_k-edges to connect these two cycles. An $HC(EPM[k] – \{e_1, e_2\})$ is constructed by adding $(k–1; 0, 1)(k; 0, 2)$ and $(k–1; 2^{k–1}–1, 1)(k; 2^k–1, 2)$ (or $(k–1; 0, 2^{k–1}–2)(k; 0, 2^k–3)$ and $(k–1; 2^{k–1}–1, 2^{k–1}–2)(k; 2^k–1, 2^k–3)$) after removing $(k–1; 0, 1)(k–1; 2^{k–1}–1, 1)$ and $(k; 0, 2)(k; 2^k–1, 2)$ (or $(k–1; 0, 2^{k–1}–2)(k–1; 2^{k–1}–1, 2^{k–1}–2)$ and $(k; 0, 2^k–3)(k; 2^k–1, 2^k–3)$).

Case 2. (Only one faulty edge in $EPM[k–1]$): Without lose of generality, we assume that e_1 is in $EPM[k–1]$. By induction hypothesis, we have an $HC(EPM[k–1] – \{e_1\})$ and two subcases are discussed as follows.

Subcase 2.1. (The other faulty edge is a L_k-edge): We first construct an $HC(T(2^k, 2^k))$ by Algorithm 1. Then, we connect the $HC(EPM[k–1] – \{e_1\})$ and the $HC(T(2^k, 2^k))$ as Case 1 to form an $HC(EPM[k] – \{e_1, e_2\})$ but the faulty L_k-edge is not one of the two chosen L_k-edges.

Subcase 2.2. (The other faulty edge is a M_k-edge): We first construct an $HC(T(2^k, 2^k) – \{e_2\})$ by Algorithm 3. Then, we connect the $HC(EPM[k–1] – \{e_1\})$ and $HC(T(2^k, 2^k) – \{e_2\})$ as Case 1 to form an $HC(EPM[k] – \{e_1, e_2\})$.

Case 3. (No faulty edges in $EPM[k–1]$): By induction hypothesis, we have an $HC(EPM[k–1])$ and three subcases are discussed as follows.

Subcase 3.1. (Two faulty edges are M_k-edges): We first construct an $HC(T(2^k, 2^k) – \{e_1, e_2\})$ by Algorithm 4. Then, we connect the $HC(EPM[k–1])$ and $HC(T(2^k, 2^k) – \{e_1, e_2\})$ as Case 1 to form an $HC(EPM[k] – \{e_1, e_2\})$.

Subcase 3.2. (One faulty edge is a L_k-edge and the other is a M_k-edge): Without lose of generality, we assume that e_1 is a L_k-edge. We first construct an $HC(T(2^k, 2^k) – \{e_1\})$ by Algorithm 3. Then, we connect the $HC(EPM[k–1])$ and $HC(T(2^k, 2^k) – \{e_1\})$ as Case 1 to form an $HC(EPM[k] – \{e_1, e_2\})$ but the faulty L_k-edge is not one of the two chosen L_k-edges.

Subcase 3.3. (Both two faulty edges are L_k-edges): If one faulty L_k-edge is incident to $(k; 0, 2)$ or $(k; 2^k–1, 2)$ and the other faulty L_k-edge is incident to $(k; 0, 2^k–3)$ or $(k; 2^k–1, 2^k–3)$, then we first apply $R_{90°}(EPM[n])$. We construct the $HC(T(2^k, 2^k))$ by Algorithm 1. Finally, we connect the $HC(EPM[k–1])$ and the $HC(T(2^k, 2^k))$ as Case 1 to form an $HC(EPM[k] – \{e_1, e_2\})$ but the faulty L_k-edge is not one of the two chosen L_k-edges.

Notice that the constructed $HC(EPM[k] – \{e_1, e_2\})$ still contains the two T_k-edges that are in the 2nd and $(2^k–1)$th rows, respectively, for further connection. ■

6 Conclusion and Future Research

It is known that $PM[n]$ is 1-Hamiltonicity [18]. Based on this result, we have shown that $EPM[n]$ is 1-Hamiltonicity. In this paper, we also showed that $EPM[n]$ is 2-edge-Hamiltonicity. This result is optimal because the node connectivity and edge connectivity of $EPM[n]$ are both four, and at most two edges can be fault. We claim that $EPM[n]$ is 2-Hamiltonicity. So, we now try to construct a Hamiltonian cycle on $EPM[n]$ with two faulty nodes and then try to show that $EPM[n]$ with one node and one edge fault is Hamiltonian. If these two extra results could be shown, we can conclude that $EPM[n]$ is 2-Hamiltonicity.

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